MACHINE LEARNING FASTAPI STREAMLIT UI 81 TESTS PASSING

Customer Churn Intelligence & Retention Optimization Platform

Complete Technical Specification, Mathematical Foundations, and Executive Business Report

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1. Executive Summary (The Project in Plain English)

In any subscription business—especially telecommunications—**churn** represents the loss of existing customers who cancel their contracts or migrate to competitors. Acquiring a brand-new customer requires expensive marketing campaigns, sales commissions, onboarding hardware, and account setup costs. Research consistently confirms that **retaining an existing subscriber costs 5 to 7 times less than acquiring a new one**.

> 💡 **The Healthcare Analogy (AI Doctor for Customer Accounts)**
>
> Think of this platform as an AI doctor for customer accounts:
>
> - **The Diagnosis:** Instead of subjective guessing, it inspects 20 concrete operational indicators (contract duration, monthly rate, tech support tickets, billing methods).
> - **True Risk Calibration:** Rather than a generic classifier score, it generates a mathematically calibrated probability (e.g., "*This customer has a genuine 73.5% chance of leaving*").
> - **Root Cause Explanation (SHAP):** It uncovers why the customer is unhappy (e.g., "*Paying $85/month for Fiber Optic without Tech Support on a Month-to-Month plan*").
> - **The Prescription:** It evaluates whether offering a $35 retention promotion yields a positive net return, protecting company funds from wasteful spending.

7,043	0.8445	-15.5%	+85.9%
ANALYZED ACCOUNTS	TEST ROC-AUC	BRIER ERROR REDUCTION	SIMULATED CAMPAIGN ROI

2. Problem Statement & Traditional Pitfalls

Traditional corporate machine learning models for customer churn fail in production due to four recurring flaws:

1. **The Arbitrary 0.5 Threshold:** Treating any customer with a score > 0.5 as "churning" ignores the asymmetric economic costs of False Positives (spending money on someone who was never leaving) vs. False Negatives (losing a high-value $1,200/year customer).
2. **Uncalibrated Classifier Scores:** Raw scores output by modern gradient boosting algorithms are ranking metrics, not real probabilities. Relying on uncalibrated scores to budget cash causes massive financial forecasting errors.
3. **Target Leakage in Pipelines:** Computing data imputation or categorical encoding across the full dataset prior to train/test partitioning contaminates the test set, creating models that look stellar in notebooks but collapse in production.
4. **Whitespace & Data Quality Traps:** The IBM Telco dataset contains 11 customer records with 0 tenure months where `TotalCharges` is stored as a blank space (`" "`). Without schema contracts, standard parsers crash or silently assign invalid values.

3. Core Project Objectives

- **Enforce Strict Data Contracts:** Programmatic ingestion with SHA256 verification and Pandera schema validation.
- **Normalized Relational Architecture:** 3NF PostgreSQL / SQLite storage for rapid analytical queries.
- **Leakage-Free Modeling:** Strict scikit-learn `ColumnTransformer` and `Pipeline` encapsulation.
- **Probability Calibration:** Platt Sigmoid Scaling to ensure model scores reflect true empirical likelihoods.
- **Cost-Sensitive Decision Boundaries:** Finding the optimal operational threshold ($\tau^* = 0.540$) using cost-benefit optimization.
- **Transparent Explainable AI (SHAP):** Global beeswarm drivers and individual customer waterfall attributions.
- **Financial Retention Simulator:** 4-way data segregation tracking Observed, Predicted, Assumed, and Simulated metrics.
- **Production Microservices:** Type-safe FastAPI REST API endpoints and a 5-tab executive Streamlit web dashboard.

4. Dataset Breakdown & Relational Schema

The platform is evaluated on the canonical **IBM Telco Customer Churn** dataset:

- **Total Customer Records:** 7,043
- **Churn Baseline:** 1,869 churners (26.5%) vs. 5,174 retained accounts (73.5%).
- **Feature Categories:**
  - *Demographics:* Gender, SeniorCitizen, Partner, Dependents.
  - *Account Details:* Tenure (months), Contract type (Month-to-month, 1-Year, 2-Year), PaperlessBilling, PaymentMethod.
  - *Services:* PhoneService, MultipleLines, InternetService (DSL, Fiber optic, None).
  - *Add-On Security & Support:* OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies.
  - *Charges:* MonthlyCharges, TotalCharges.

5. Leakage-Safe Machine Learning & Benchmark Results

All data preprocessing (imputation, scaling, and one-hot encoding) is packaged inside a scikit-learn `Pipeline`. Models were evaluated using 5-Fold Stratified Cross-Validation on the training fold (80%), with final reporting on 1,409 unobserved held-out accounts (20% partition):

Model Architecture	PR-AUC	ROC-AUC	F1 Score	Precision	Recall	Brier Score	Status
Logistic Regression	0.6325	0.8415	0.6136	0.5043	0.7834	0.1687	Baseline
Random Forest	0.6532	0.8409	0.6358	0.5325	0.7888	0.1609	Ensemble
XGBoost Classifier	**0.6623**	**0.8445**	**0.6372**	**0.5333**	**0.7914**	**0.1610**	⭐ **Top Performer**

6. Probability Calibration & Cost-Benefit Threshold Tuning

Platt Sigmoid Probability Calibration

Raw XGBoost outputs were calibrated using 5-fold cross-validated Platt scaling (`CalibratedClassifierCV`):

- **Brier Score:** Reduced from 0.1610 to **0.1361 (-15.46% error reduction)**.
- **Log Loss:** Dropped from 0.5283 to **0.4182 (-20.84% error reduction)**.
- **Expected Calibration Error (ECE):** Improved by **68.3%**, aligning scores directly with observed churn frequencies.

Cost-Benefit Decision Optimization

Assuming an intervention offer cost of $35.00, an assumed save rate of 22%, and a customer lifetime value (CLV) of $450.00:

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Theoretical Bayes Threshold: tau* = Offer_Cost / (Save_Rate x CLV) =
35 / (0.22 x 450) = 0.3535
Cross-Validation Optimal Threshold: tau* = 0.540
```

Strategy	Decision Threshold	Targeted Accounts	Net Saved Revenue ($)	Efficiency Lift
Standard Naive Baseline	0.500	555	$9,246.00	Baseline
Cost-Optimal Boundary	**0.540**	**509**	**$9,365.00**	**+$119.00 Lift (46 fewer wasted calls)**

7. Explainable AI: Global & Local SHAP Attributions

TreeSHAP explains both macroscopic company-wide patterns and individual subscriber cases:

- **Top Churn Hazards (Global):** Month-to-Month contracts, high monthly subscription rates, Fiber Optic connection without bundled technical assistance, and payment via Electronic Check.
- **Top Protective Factors:** Two-Year contracts, prolonged account tenure (> 24 months), bundled Online Security, and automated bank transfers.
- **Frontline Actionability:** Customer service reps receive waterfall attributions showing the exact push factors for any specific subscriber during live retention calls.

8. Financial Retention Budget Simulator

To prevent counterfactual assumptions from being confused with recorded historical facts, all simulation tables enforce strict four-way column categorization:

- `[OBSERVED DATA]`: Historic facts (tenure, billing rate, contract type).
- `[MODEL PREDICTION]`: Calibrated probability and risk tier.
- `[HYPOTHETICAL ASSUMPTION]`: Offer discount cost, campaign budget, assumed retention success rate.
- `[SIMULATED OUTCOME]`: Expected gross revenue saved, net profit, campaign ROI, and target ranking.

> 💰 **Empirical Campaign Outcomes ($10,000 Budget Simulation)**
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> - **Budget Utilization:** $9,975.00 (285 high-risk customers prioritized, 99.8% budget used).
> - **Expected Accounts Saved:** 41.2 accounts.
> - **Expected Gross Saved Revenue:** $18,540.72.
> - **Expected Net Saved Revenue: $8,565.72**.
> - **Simulated Campaign ROI: +85.9%**.

9. Production Serving & User Interfaces

FastAPI Prediction Microservice (Port 8000)

- `POST /predict`: Real-time scoring for single customer JSON payloads.
- `POST /predict/batch`: High-throughput batch inference (up to 500 accounts simultaneously).
- `GET /health`: Microservice readiness, active model metadata, and database heartbeat.
- `GET /analytics/summary`: Live relational database aggregate metrics.

Streamlit Executive Dashboard (Port 8501)

- **Tab 1: 📊 Executive & Dataset KPIs:** High-level revenue at risk, churn distribution, and cohort decay curves.
- **Tab 2: 🤖 Model Benchmarks & Calibration:** ROC-AUC curves, reliability diagrams, and threshold sweeps.
- **Tab 3: 🎯 Customer Risk Scoring & Sandbox:** Searchable customer roster and live interactive single-account scoring.
- **Tab 4: 🔍 Explainable AI (SHAP):** Beeswarm summary plots and local waterfall attributions.
- **Tab 5: 💰 Retention Budget Simulator:** Real-time budget sliders, efficiency frontiers, and what-if scenario testing.

10. Verification & Quality Assurance

The system is protected by **81 automated unit and integration tests** executed via Pytest:

- Data ingestion, Pandera schema checks, and whitespace handling.
- PostgreSQL and SQLite relational database schema creation and queries.
- Pipeline leakage checks (asserting transformers only fit on training folds).
- Calibration Brier score calculations and threshold utility sweeps.
- Retention budget math, diminishing returns ranking, and ROI formulas.
- FastAPI endpoint schema validation, edge case rejections, and batch scoring.
- Dashboard analytical helper calculations and Plotly component rendering.
